

NEO-6 series

Versatile u-blox 6 GPS modules

Highlights

- Miniature package
- UART, USB, DDC (I²C compliant) and SPI interfaces
- Available in Crystal and TCXO versions
- Onboard RTC crystal for faster warm and hot starts
- 1.8 V and 3.0 V variants



NEO-6:
12.2 x 16.0 x 2.4 mm

Features

- u-blox 6 position engine:
 - Navigate down to -162 dBm and -148 dBm coldstart
 - Faster acquisition with AssistNow Autonomous
 - Configurable power management
 - Hybrid GPS/SBAS engine (WAAS, EGNOS, MSAS)
 - Anti-jamming technology
- Simple integration with u-blox wireless modules
- A-GPS: AssistNow Online and AssistNow Offline services, OMA SUPL compliant
- Backward compatible (hardware and firmware); easy migration from NEO-5 family or NEO-4S
- Based on GPS chips qualified according to AEC-Q100
- Manufactured in ISO/TS 16949 certified sites
- LCC package for reliable and cost effective manufacturing
- Operating temperature range: -40°C to 85°C

Product description

The NEO-6 module series brings the high performance of the u-blox6 position engine to the miniature NEO form factor. u-blox6 has been designed with low power consumption and low costs in mind. Intelligent power management is a breakthrough for low-power applications. These receivers combine a high level of integration capability with flexible connectivity options in a small package. This makes them perfectly suited for mass-market end products with strict size and cost requirements. The DDC interface provides connectivity and enables synergies with u-blox LEON and LISA wireless modules.

All NEO-6 modules are based on GPS chips qualified according to AEC-Q100 and are manufactured in ISO/TS 16949 certified sites. Qualification tests on NEO-6 modules are performed as stipulated in the ISO16750 standard: "Road vehicles – Environmental conditions and testing for electrical and electronic equipment".

Product selector

Model	Type	Supply	Interfaces				Features						
	Standalone GPS Capture & Process Timing & Raw Data Dead Reckoning	1.75 V – 2.0 V 2.7 V – 3.6 V	UART	USB	SPI	DDC (I2C compliant)	Programmable (Flash) FW update	TCXO (KickStart)	RTC crystal	Antenna supply and supervisor	Configuration pins	Timepulse	External interrupt / Wakeup
NEO-6G	•	•	•	•	•	•	•	•	•	○	3	1	•
NEO-6Q	•	•	•	•	•	•	•	•	•	○	3	1	•
NEO-6M	•	•	•	•	•	•			•	○	3	1	•

◦ = requires external components and integration on application processor

Receiver performance data

Receiver type	50-channel u-blox 6 engine GPS L1 C/A code SBAS: WAAS, EGNOS, MSAS		
Navigation update rate	up to 5 Hz		
Accuracy ¹	Position	2.5 m CEP	
	SBAS	2.0 m CEP	
Acquisition ¹		NEO-6G/Q	NEO-6M
	Cold starts:	26 s	27 s
	Aided starts ² :	1 s	< 3 s
	Hot starts:	1 s	1 s
Sensitivity ³		NEO-6G/Q	NEO-6M
	Tracking:	-162 dBm	-162 dBm
	Cold starts:	-148 dBm	-147 dBm
	Hot starts:	-157 dBm	-156 dBm

¹ All SV @ -130 dBm

² Dependent on aiding data connection speed and latency

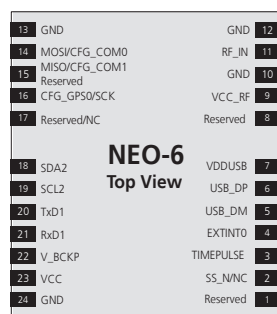
³ Demonstrated with a good active antenna

Electrical data

Power supply	2.7 V – 3.6 V (NEO-6Q/6M) 1.75 V – 2.0 V (NEO-6G)
Power consumption	117 mW @ 3.0 V (continuous) 33 mW @ 3.0 V Power Save Mode (1 Hz) 72 mW @ 1.8 V (continuous) 22 mW @ 1.8 V Power Save Mode (1 Hz)
Backup power	1.4 V – 3.6 V, 22 µA
Supported antennas	Active and passive

Package

Dimensions	LCC (Leadless Chip Carrier), surface mount package: 12.2 x 16.0 x 2.4 mm
Weight	1.6 g
Pinout	



Legal Notice

u-blox reserves all rights to this document and the information contained herein. Products, names, logos and designs described herein may in whole or in part be subject to intellectual property rights. Reproduction, use, modification or disclosure to third parties of this document or any part thereof without the express permission of u-blox is strictly prohibited.

The information contained herein is provided "as is". No warranty of any kind, either express or implied, is made in relation to the accuracy, reliability, fitness for a particular purpose or content of this document. This document may be revised by u-blox at any time. For most recent documents, please visit www.u-blox.com.

Copyright © 2011, u-blox AG

Specification applies to FW 7.01

Interfaces

Serial interfaces	1 UART 1 USB V2.0 full speed 12 Mbit/s 1 DDC (I ² C compliant) 1 SPI	
Digital I/O	Configurable timepulse 1 EXTINT input for Wakeup	
Serial and I/O	Voltages	2.7 – 3.6 V (NEO-6Q/6M) 1.75 – 2.0 V (NEO-6G)
Timepulse	Configurable	0.25 Hz to 1 kHz
Protocols	NMEA, UBX binary, RTCM	

Environmental data

Operating temp.	-40° C to 85° C
Storage temp.	-40° C to 85° C
RoHS compliant (lead-free)	

Support products

u-blox 6 Evaluation Kits:

Easy-to-use kits to get familiar with u-blox 6 positioning technology, evaluate functionality, and visualize GPS performance.

EVK-6H:	u-blox 6 Evaluation Kit with TCXO, suitable for NEO-6G, NEO-6Q
EVK-6P:	u-blox 6 Evaluation Kit with crystal, suitable for NEO-6M

Ordering information

NEO-6G-0	ROM-based u-blox 6 GPS Module 1.8 V with TCXO
NEO-6M-0	ROM-based u-blox 6 GPS Module with Crystal
NEO-6Q-0	ROM-based u-blox 6 GPS Module with TCXO

Available as samples and tape on reel (250 pieces)